

Dynamics of Forced Particles

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under

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1 Introduction

In the natural world, systems evolve over time — from falling objects to growing populations, and modelling this evolution leads us to differential equations. Based on the number of independent variables, differential equations are classified into two types:

- **Ordinary Differential Equations (ODEs)**: involve functions of a single variable (e.g., time).
- **Partial Differential Equations (PDEs)**: involve functions of multiple variables (e.g., like temperature changing over time and space).

ODEs are commonly used in modelling mechanical systems such as the motion of projectiles and pendulums. PDEs, on the other hand, describe distributed systems, such as heat conduction in solids or fluid dynamics.

However in reality, not all differential equations can be solved exactly. Even seemingly simple real-world equations (like that of a pendulum with air resistance) are too complex for "neat" solutions. Instead, we might explore their behaviour using simulations, visualizations, or approximate numeric methods. This practical limitation doesn't reduce their importance.

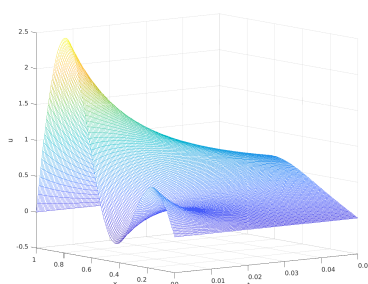


Figure 1: Solution of Heat Equation

2 The Heat Equation and the Need for Decomposition

Let's start PDEs just like Joseph Fourier began with. He was working on finding solution for the famous **heat equation**, which models how temperature evolves in a rod over time. For PDEs like the heat equation for 1D rod,

$$\frac{\partial T(x, t)}{\partial t} = \alpha \frac{\partial^2 T(x, t)}{\partial x^2}$$

we require sophisticated tools. While solving it directly for arbitrary initial temperature profiles is difficult, mathematicians found that for specific wave-like profiles — say, cosine or sine waves — solutions become surprisingly tractable. Consider the boundary

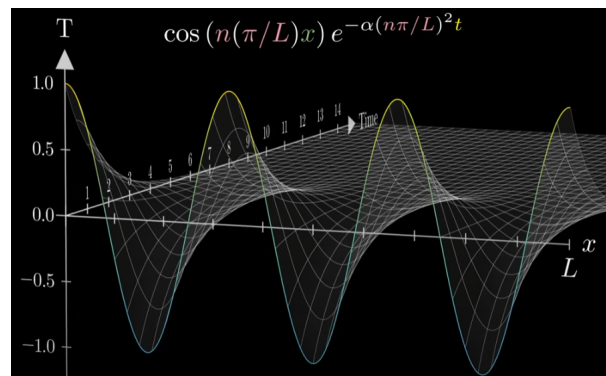


Figure 2: Heat equation solution for given boundary conditions

conditions,

$$\frac{\partial T(0, t)}{\partial t} = \frac{\partial T(L, t)}{\partial x} = 0$$

A powerful trick emerges here: since the heat equation is **linear**, combinations of simple wave solutions (each decaying over time) can be added together to build more complex solutions. If you can decompose an initial temperature profile into a sum of sines and cosines, each of these can evolve individually, and their sum gives the final answer.

$$T(x, t) = \sum_{n=-\infty}^{\infty} \cos(n(\pi/l)x) e^{-\alpha(n\pi/l)^2 t}$$

But what if your initial temperature distribution isn't a wave at all — say, it's a step function? Fourier's insight was that solutions for simple sine wave initial conditions decay exponentially, and any initial condition can be represented as a sum of such sine waves—a concept that led to Fourier series.

Why sinusoids especially?

Sinusoids are used to decompose signals because they are:

- **Eigenfunctions of LTI systems** (linearity and time-invariance of the system, together ensure that sinusoidal inputs produce **sinusoidal outputs of the same frequency**. Convolution with an impulse response preserves the exponential form of the input — scaling it by a frequency-dependent complex gain $H(j\omega)$.
- **Orthogonal** and complete (mathematically robust basis). On the interval $[0, 2\pi]$, the sine and cosine functions are orthogonal:

$$\int_{-\pi/\omega}^{\pi/\omega} \cos(j\omega t) \cos(k\omega t) dt = \begin{cases} 0 & \text{if } j \neq k \\ \frac{\pi}{\omega} & \text{if } j = k \end{cases}$$

This orthogonality ensures that when projecting a function onto the sine and cosine



bases, the coefficients are computed independently, allowing for efficient decomposition and reconstruction.

- Connected to geometry via complex exponentials (the epicycle representation).

3 Summing Waves to solve everything

Fourier series allow us to express any complex, messy or even discontinuous function as an infinite sum of sines and cosines. Mathematically,

$$f(x) = a_0 + \sum_{n=1}^{\infty} (a_n \cos(nx) + b_n \sin(nx))$$

Each coefficient a_n, b_n tells us how much of each frequency contributes to building up the function. In complex form, the Fourier series is:

$$f(t) = \sum_{n=-\infty}^{\infty} c_n e^{2\pi i n t}$$

with coefficients computed by:

$$c_n = \int_0^1 f(t) e^{-2\pi i n t} dt$$

These coefficients encode both amplitude and phase of the harmonic components of the signal.

Geometrically, each term corresponds to a rotating vector (epicycle) in the complex plane.

$$e^{it} = \cos(t) + i \sin(t)$$

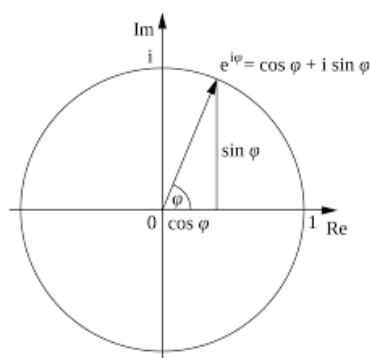


Figure 3: Euler's formula on Complex Plane

Summing these vectors reconstructs the original function as a path traced by the end



point of the sum which underpins applications in animation, signal synthesis, and systems modelling .



Figure 4: Epicyclic representation for Fourier series

4 Non-linear Vibrating Systems - Model Description

The systems we've studied so far are linear or linearized-like heat conduction or small vibrations. However, in many mechanical and structural systems, especially with lightweight components or flexible structures, vibratory motion is not purely linear. Non-linearities arise due to large deformations, geometric coupling between modes, or non-linear restoring forces (e.g., αx^3 spring terms). These effects are especially significant in aerospace components, MEMS devices, and turbine blades, where accurate modelling of oscillatory behaviour is critical.

Consider a general second-order mechanical system with N degrees of freedom (DOFs):

$$\mathbf{M}\ddot{\mathbf{x}}(t) + \mathbf{C}\dot{\mathbf{x}}(t) + \mathbf{K}\mathbf{x}(t) + \mathbf{f}_{nl}(\mathbf{x}(t)) = \mathbf{F}_{ext}(t)$$

where:

- $\mathbf{x}(t) \in R^N$ is the displacement vector
- $\mathbf{M}, \mathbf{C}, \mathbf{K}$ are the mass, damping, and stiffness matrices
- $\mathbf{f}_{nl}$ represents the non-linear internal forces (e.g., geometric or material non-linearities)
- $\mathbf{f}_{ext}(t)$ is the external forcing, often periodic

Solving these non-linear ODEs analytically is challenging. Solving such systems directly over long time durations using time-stepping schemes (e.g., Runge–Kutta, Newmark–Beta) becomes computationally expensive and is also a waste of storage. Some FEM models numerically compute for thousands of DOFs out of which only very few would have been non-linear. A hybrid semi-analytical approach involving Harmonic Balance Method (HBM) provides a powerful way to approximate periodic solutions.

1. **Harmonic Balance Method (HBM):** To approximate periodic steady-state solutions using truncated Fourier series.



2. **Component Mode Synthesis (CMS)**: To reduce the number of DOFs by exploiting the structure's modal behaviour (retaining only dominant normal modes and constraint modes)
3. **Newton-Raphson Iteration**: Since the resulting algebraic system is non-linear, it is solved iteratively using the Newton-Raphson method.

4.1 Governing Equation of the System

The study focuses on a vibrating mechanical system with internal resonance. A prototypical example is a mass-spring-damper system with cubic stiffness:

$$\ddot{x}(t) + 2\zeta\omega_0\dot{x}(t) + \omega_0^2x(t) + \alpha x^3(t) = F \cos(\omega t)$$

This is known as the **Duffing oscillator**. The cubic term αx^3 induces hardening or softening behaviour depending on the sign of α .

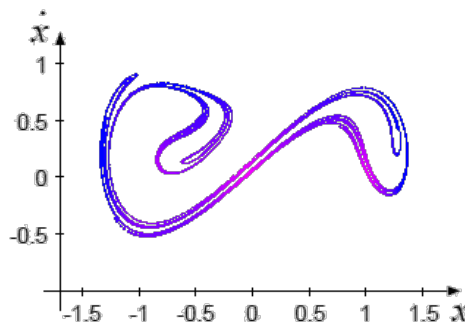


Figure 5: Periodic change of the chaotic attractor of the Duffing oscillator

In the general setting, such non-linearities can couple modes of vibration, particularly if two natural frequencies $\omega_i \approx \omega_j$, leading to **internal resonance**. A special case is the 1:1 resonance, where two modes exchange energy, requiring accurate modelling of both simultaneously.

4.2 Model Reduction using Component Mode Synthesis (CMS)

Concept:

- Divide the structure into substructures.
- Use modal coordinates for slave substructure (linear).
- Use physical coordinates for non-linear substructure.

Two types of DOFs:



1. **Interior DOFs (i)**: inside the substructure. We express internal DOFs x_i using a set of reduced set of modal amplitudes u_s and constraint contributions from x_j .
2. **Junction DOFs (j)**: interface DOFs between substructures. Here, full physical coordinates are retained x_j , so that we can connect this substructure to others.

The displacement vector is transformed using:

$$\begin{bmatrix} x_i \\ x_j \end{bmatrix} = \underbrace{\begin{bmatrix} \Phi_R & \Psi \\ 0 & I \end{bmatrix}}_T \begin{bmatrix} u_s \\ x_j \end{bmatrix}$$

Where :

Φ_R - Retained eigenmodes of internal DOFs

Ψ - Constraint(static) modes model the static deformation of internal DOFs x_i due to unit displacement at junctions x_j .

This reduces the system to a much smaller set of generalized coordinates u_s , while still accounting for the coupling through junction DOFs x_j .

5 Solving with Harmonic Balance Method

5.1 Procedure

5.1.1 Truncated Fourier Series

In the Harmonic Balance Method (HBM), the response vector $x_k(t)$, external force vector $F_k(t)$, and non-linear function components $f_k(x, \dot{x})$, $f_{pk}(x, \dot{x})$ for each degree of freedom (DOF) are expressed as truncated Fourier series (with n harmonics) as:

$$x_k(t) = \bar{x}_{k0} + \sum_{l=1}^n \{ \bar{x}_{kl}^c \cos(l\omega t) + \bar{x}_{kl}^s \sin(l\omega t) \} \quad (1)$$

$$F_k(t) = \bar{F}_{k0} + \sum_{l=1}^n \{ \bar{F}_{kl}^c \cos(l\omega t) + \bar{F}_{kl}^s \sin(l\omega t) \} \quad (2)$$

$$f_k(x, \dot{x}) = \bar{f}_{k0} + \sum_{l=1}^n \{ \bar{f}_{kl}^c \cos(l\omega t) + \bar{f}_{kl}^s \sin(l\omega t) \} \quad (3)$$

$$f_{pk}(x, \dot{x}) = \bar{f}_{pk0} + \sum_{l=1}^n \{ \bar{f}_{pkl}^c \cos(l\omega t) + \bar{f}_{pkl}^s \sin(l\omega t) \} \quad (4)$$

where:

- $k = 1, 2, \dots, N$: index of the DOF



- $l = 1, 2, \dots, n$: harmonic index
- ω : fundamental forcing frequency
- $\bar{x}_{k0}, \bar{F}_{k0}, \bar{f}_{k0}$: mean (DC) component
- $\bar{x}_{kl}^c, \bar{F}_{kl}^c, \bar{f}_{kl}^c$: cosine coefficients
- $\bar{x}_{kl}^s, \bar{F}_{kl}^s, \bar{f}_{kl}^s$: sine coefficients

The vector form of the truncated Fourier series coefficients for the k^{th} DOF is:

$$\tilde{x}_k = \begin{bmatrix} \bar{x}_{k0} & \bar{x}_{k1}^c & \bar{x}_{k1}^s & \bar{x}_{k2}^c & \bar{x}_{k2}^s & \cdots & \bar{x}_{kn}^c & \bar{x}_{kn}^s \end{bmatrix}^T$$

This is a column vector of length $2n + 1$ for each DOF, with 1 constant term, n cosine terms, and n sine terms. If there are N DOFs, the total number of unknowns in the Fourier space is $N(2n + 1)$.

Truncating the Fourier series at n harmonics introduces a tradeoff:

- A low value of n results in a coarse approximation that may miss important non-linear behaviour or harmonic content.
- A high value of n increases accuracy but significantly raises the computational cost (since the system size grows as $N(2n + 1)$).

After substituting the Fourier expansions of all terms into the general equation, we get

$$\begin{bmatrix} \mathbf{E}_{11} & \mathbf{E}_{12} & \cdots & \mathbf{E}_{1N} \\ \mathbf{E}_{21} & \mathbf{E}_{22} & \cdots & \mathbf{E}_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{E}_{N1} & \mathbf{E}_{N2} & \cdots & \mathbf{E}_{NN} \end{bmatrix} \begin{bmatrix} \tilde{\mathbf{x}}_1 \\ \tilde{\mathbf{x}}_2 \\ \vdots \\ \tilde{\mathbf{x}}_N \end{bmatrix} + \begin{bmatrix} \tilde{\mathbf{f}}_{p1} \\ \tilde{\mathbf{f}}_{p2} \\ \vdots \\ \tilde{\mathbf{f}}_{pN} \end{bmatrix} + \begin{bmatrix} \tilde{\mathbf{f}}_1 \\ \tilde{\mathbf{f}}_2 \\ \vdots \\ \tilde{\mathbf{f}}_N \end{bmatrix} - \begin{bmatrix} \tilde{\mathbf{F}}_1 \\ \tilde{\mathbf{F}}_2 \\ \vdots \\ \tilde{\mathbf{F}}_N \end{bmatrix} = \mathbf{R}(\alpha)$$

Here, $R(\gamma)$ is the residual error due to Fourier series truncation (since the Fourier series is infinite in theory, but we truncate it to n terms).

The matrix $\tilde{Y}$ is assembled block-wise from matrices E_{ij} defined as:

$$E_{ij} = \begin{bmatrix} k_{ij} & 0 & 0 & \cdots & 0 \\ 0 & k_{ij} - \omega^2 m_{ij} & \omega c_{ij} & \cdots & 0 \\ 0 & -\omega c_{ij} & k_{ij} - \omega^2 m_{ij} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & k_{ij} - n^2 \omega^2 m_{ij} \end{bmatrix}$$

Each E_{ij} is of size $(2n + 1) \times (2n + 1)$, and $\tilde{Y}$ is of size $N(2n + 1) \times N(2n + 1)$.

Our aim is to transform the full system of differential equations into a **non-linear algebraic system** in terms of the Fourier coefficients.



5.1.2 Galerkin Projection and Residuals

To transform the original non-linear differential equations into a solvable algebraic form, we apply the **Galerkin method** — a type of *weighted residual method* — using the Fourier basis functions as test functions.

The idea is simple: since we are approximating the solution using a truncated Fourier series, we want the residual (the mismatch between the true dynamics and the approximate model) to be *orthogonal* to all basis functions used in the approximation. This ensures that the error is minimized across the entire function space.

Mathematically: Let $R(t)$ denote the residual of the governing differential equation after substituting the Fourier expansions for $x(t)$, $f(x, \dot{x})$, $f_p(x, \dot{x})$, and $F(t)$.

We require the residual to satisfy:

$$\int_0^T R(t) \cdot \phi_n(t) dt = 0 \quad \forall \phi_n(t) \in \{1, \cos(n\omega t), \sin(n\omega t)\} \quad (5)$$

This integral condition ensures that each harmonic component of the residual is individually minimized.

Alternatively, in vector notation, this is often expressed as:

$$\sum_{n=1}^m \int_0^{2\pi/\omega} R(\alpha) \begin{bmatrix} 1 \\ \cos(n\omega t) \\ \sin(n\omega t) \end{bmatrix} dt = 0 \quad (6)$$

The result of this projection is a residual vector $R(\tilde{x}_a)$ in the frequency domain. The Harmonic Balance Method then seeks the steady-state solution that drives this residual vector to zero:

$$R(\tilde{x}_a) = 0$$

Which gives the system of non-linear algebraic equations:

$$\tilde{Y}\tilde{x}_a + \tilde{f}_p + \tilde{f} - \tilde{F} = 0 \quad (7)$$

Here:

- $\tilde{x}_a$: stacked vector of Fourier coefficients for all degrees of freedom (DOFs)
- $\tilde{Y}$: system matrix in Fourier space (assembled from element matrices E_{ij})
- $\tilde{f}, \tilde{f}_p, \tilde{F}$: Fourier coefficients of the non-linear forces and external excitation

This final equation (7) represents $N(2n+1)$ non-linear algebraic equations in $N(2n+1)$ unknowns — one for each Fourier coefficient of each DOF — ensuring the response



balances all harmonics in the system.

Solving Strategy: The process involves the following steps:

1. Start with an initial guess for $\tilde{x}_a$
2. Use the inverse Fourier transform to obtain the time-domain displacement $x(t)$
3. Compute $\dot{x}(t)$ using numerical differentiation
4. Evaluate non-linear functions $f(x, \dot{x})$, $f_p(x, \dot{x})$, and external force $F(t)$ at discrete time points
5. Apply the Discrete Fourier Transform (DFT) to convert these functions back into their frequency-domain representations $\tilde{f}$, $\tilde{f}_p$, $\tilde{F}$
6. Construct the residual $R(\tilde{x}_a) = \tilde{Y}\tilde{x}_a + \tilde{f}_p + \tilde{f} - \tilde{F}$
7. Use Newton–Raphson iteration to update $\tilde{x}_a$ and repeat until convergence

5.1.3 Newton-Raphson Iteration

The Newton-Raphson method is applied to solve for $\tilde{x}_a$:

$$J = \frac{\partial(Y\tilde{x}_a + \tilde{f}_p + \tilde{f} - \tilde{F})}{\partial\tilde{x}_a}$$

Here, J is the Jacobian used to iteratively update the coefficients until convergence. The DFT is used to compute $\tilde{f}_p$, $\tilde{f}$, and $\tilde{F}$ from time-domain evaluations of $x(t)$, $\dot{x}(t)$, which are used to evaluate the nonlinear forces.

5.1.4 Choosing the Number of Harmonics n

The accuracy of the Harmonic Balance Method (HBM) depends on the number of harmonics n used in the truncated Fourier series expansion. Choosing an appropriate value of n is crucial for ensuring a good approximation without unnecessary computational overhead.

The decision is typically guided by the following principles:

- **Prior Knowledge:** If the physical system is known to contain only a few dominant frequencies (e.g., near-resonant behaviour), a low n may suffice.
- **Convergence Study:** Begin with a small value of n and incrementally increase it. Stop when the change in the solution (or the norm of the residual - error) becomes negligible.



- **Energy-Based Criterion (Parseval's Theorem):** Analyze how much of the signal's total energy is captured by the truncated series. Parseval's theorem relates the total energy of a signal to the sum of the squares of its Fourier coefficients:

$$x_n(t) = a_0 + \sum_{k=1}^n (a_k \cos(k\omega t) + b_k \sin(k\omega t))$$

The energy contained in the truncated series up to n harmonics is:

$$E_n = \frac{1}{2}a_0^2 + \sum_{k=1}^n (a_k^2 + b_k^2)$$

If the total energy of the signal is:

$$E_{\text{total}} = \frac{1}{2}a_0^2 + \sum_{k=1}^{\infty} (a_k^2 + b_k^2)$$

Then the **fraction of captured energy** is:

$$\eta_n = \frac{E_n}{E_{\text{total}}}$$

When $\eta_n \approx 1$ (e.g., $> 99.5\%$), it indicates that the higher harmonics contribute insignificantly, and truncation at n is acceptable.

This approach provides a **quantitative measure** for deciding the truncation point and ensures that the approximation retains the dominant spectral content of the original system.

5.2 Norms and Their Role in Convergence

In the Harmonic Balance Method (HBM), norms are used to monitor both convergence of the solution and residual error. Specifically:

- **Infinity Norm (L^∞)** of the update vector $\Delta\tilde{x}_a$:

$$\|\Delta\tilde{x}_a\|_\infty = \max_{1 \leq i \leq n} |x_i|$$

This captures the maximum absolute change in any component of the solution. A condition such as $\|\Delta\tilde{x}_a\|_\infty < \varepsilon$ ensures that no single Fourier coefficient is changing significantly, indicating stabilization of the iteration.



- **2-Norm (L^2)** of the residual vector $R(\tilde{x}_a)$:

$$\|R(\tilde{x}_a)\|_2 = \left(\sum_{i=1}^n |R_i|^2 \right)^{1/2}$$

This measures the overall error in satisfying the algebraic system. A threshold $\|R(\tilde{x}_a)\|_2 < \delta$ ensures that the system is globally approaching equilibrium.

What are the other norms?

- **L^0 Norm (Pseudo-Norm):**

$$\|x\|_0 = \text{number of non-zero elements in } x$$

This counts how many coefficients are non-zero. It's not a true norm because it doesn't satisfy all norm properties (e.g., homogeneity). It offers no gradient information or smooth behaviour, which makes it unsuitable for convergence monitoring.

- **L^1 Norm (Manhattan Norm):**

$$\|x\|_1 = \sum_{i=1}^n |x_i|$$

The L^1 norm measures total absolute deviation. It is less sensitive to large individual components than the L^∞ norm and doesn't capture overall RMS error as directly as the L^2 norm. It's often used in robust regression but is less effective for convergence control in iterative solvers.

Why only L^∞ and L^2 ? The combination of these norms provides complementary convergence checks:

- L^∞ ensures *local stability*, by bounding the largest individual update.
- L^2 ensures *global accuracy*, by minimizing the mean residual error.

5.3 Advantages of using HBM

- Captures steady-state behaviour of non-linear oscillations.
- Extensible to systems with many degrees of freedom (via modal truncation).
- Forms the basis for continuation techniques and bifurcation analysis.



6 Chaos Theory - Introduction via Dynamical Systems

In non-linear systems, it is often impossible to obtain closed-form solutions to the governing differential equations. Even when solutions exist, they may be too complex to yield meaningful insight. Thus, rather than pursuing exact solutions, we aim to understand the *qualitative behaviour* of such systems — such as their long-term dynamics, stability, and the potential for chaos — by analyzing the structure of the governing equations.

6.1 Fixed Points and Their Importance

A fixed point (or equilibrium point) is a point $\mathbf{x}^*$ where the system remains unchanged:

$$\dot{\mathbf{x}} = f(\mathbf{x}) = 0$$

At a fixed point, the system has no dynamics — any solution that starts at $\mathbf{x}^*$ stays there for all time.

6.2 Stability of Fixed Points and Linearization

To determine the stability of fixed points, we linearize the system near $\mathbf{x}^*$:

$$\dot{\mathbf{x}} = f(\mathbf{x}) \approx f(\mathbf{x}^*) + J(\mathbf{x}^*)(\mathbf{x} - \mathbf{x}^*) = J(\mathbf{x} - \mathbf{x}^*)$$

where J is the Jacobian matrix $\partial f / \partial x$ evaluated at $\mathbf{x}^*$. The behaviour near a fixed point is governed by the eigenvalues λ_i of J :

- If all $\text{Re}(\lambda_i) < 0$: the fixed point is locally stable (sink).
- If any $\text{Re}(\lambda_i) > 0$: the fixed point is unstable (source or saddle).
- If all $\text{Re}(\lambda_i) = 0$: linearization is inconclusive.

Limitations of Classical Stability Analysis

- It applies only **locally**, near fixed points.
- It cannot capture global features like long-term trajectory divergence.
- It fails to detect chaos, strange attractors, or sensitive dependence on initial conditions.

Hence, even though a system is described by differential equations, local eigenvalue analysis is insufficient to predict long-term behaviour when non-linearities are strong or trajectories evolve far from equilibria.



6.3 Phase Portraits and Qualitative Analysis

A phase portrait represents the directional behaviour of a system of ordinary differential equations (ODEs). It is a graphical representation of trajectories in the phase space. Even without solving the system, one can deduce the presence of an attractor, a repeller or limit cycle for the chosen parameter value. For linear systems, eigenvectors of J point along the straight-line solution trajectories, and eigenvalues indicate contraction or expansion along those directions. The phase portrait can indicate the stability of the system.

Unstable	Most of the system's solutions tend towards ∞ over time
Asymptotically stable	All of the system's solutions tend to 0 over time
Neutrally stable	None of the system's solutions tend towards ∞ over time, but most solutions do not tend towards 0 either

Figure 6: Table: Stability interpretations from phase portrait visualisation

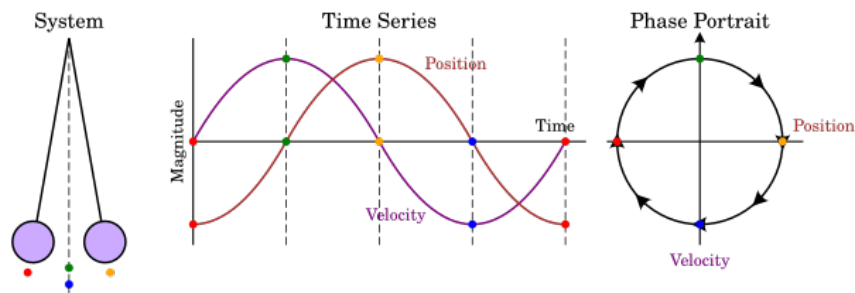


Figure 7: Phase portrait of a simple pendulum

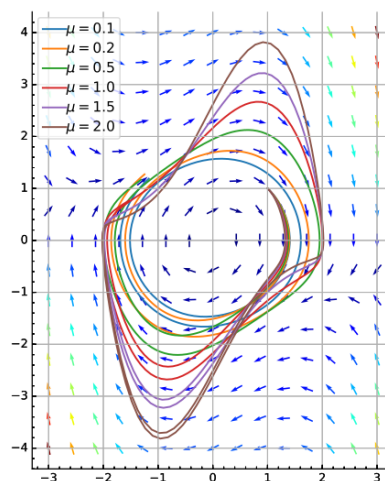


Figure 8: Van der Pol's oscillator phase portrait, $\frac{d^2y}{dt^2} + \mu(y^2 - 1)\frac{dy}{dt} + y = 0$.



6.4 When is a System Chaotic?

Though in common usage, "chaos" refers to a state of "disorderedness", in mathematics there is no universally accepted definition for it. According to Robert Devaney, a dynamic system is said to exhibit *chaos* if it satisfies the following:

1. **Sensitivity to Initial Conditions:** Small differences in initial conditions yield diverging outcomes.
2. **Non-periodic:** The system never exactly repeats its state.
3. **Topological Mixing:** Future states of the system are intertwined, like stirring dye in fluid. Intuitively, if a map is topologically transitive then given a point x and a region V , there exists a point y near x whose orbit passes through V . This implies that **it is impossible to decompose the system into two open sets.**

6.5 How Can We Measure Chaos?

Dynamical systems—especially non-linear ones—are often governed by deterministic differential equations. Yet, their long-term behaviour can be unpredictable, aperiodic, or chaotic. To understand such systems, conventional tools like linear stability analysis fall short. This motivates the use of Lyapunov exponents, which offer a global, quantitative framework and Poincaré sections for qualitative understanding and characterisation of system's stability and sensitivity.

6.5.1 Lyapunov Exponent and Stability Interpretation

The Lyapunov exponent is a quantitative measure of the rate at which nearby trajectories of a dynamical system diverge or converge. Given an initial infinitesimal separation δ_0 , the separation at time t grows as:

$$|\delta(t)| \approx e^{\lambda t} |\delta_0|$$

where λ is the Lyapunov exponent.

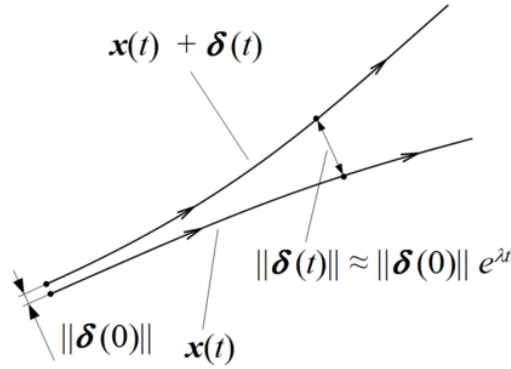


Figure 9: Explanation of Lyapunov exponent

In an n -dimensional phase space, there exists a *spectrum* of n Lyapunov exponents, each corresponding to a direction in phase space. The **maximum Lyapunov exponent** (MLE), $\lambda_{\max}$, governs the dominant rate of separation and is commonly used to assess system predictability. The *maximum Lyapunov exponent*, denoted as $\lambda_{\max}$, is defined as:

$$\lambda_{\max} = \lim_{T \rightarrow \infty} \frac{1}{T} \log |\lambda_i^T|$$

where λ_i are the eigenvalues of the Jacobian matrix J evaluated at the equilibrium points of the considered (possibly fractional-order) dynamical system. Most initial separation vectors contain a component along the direction of the MLE, causing it to dominate over time due to exponential growth.

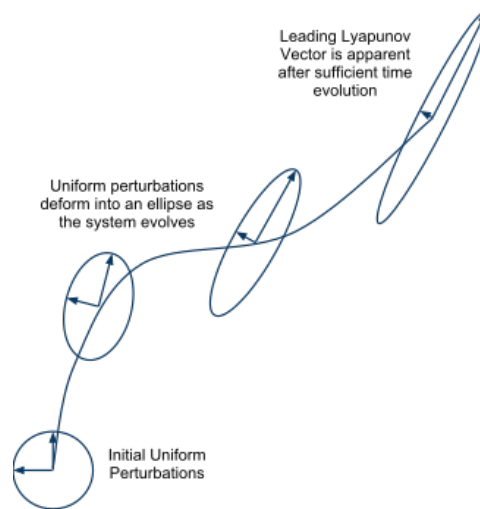


Figure 10: Leading Lyapunov Vector



Interpretation of $\lambda_{\max}$

- **If $\lambda_{\max} < 0$:** The system trajectories converge to a stable fixed point (attractor). The system is stable.
- **If $\lambda_{\max} = 0$:** The system evolves on a limit cycle or a torus. The system is neutrally stable (Lyapunov stable). Trajectories stay nearby- *periodic or quasi-periodic*.
- **If $\lambda_{\max} > 0$:** The system exhibits chaotic behaviour; nearby trajectories diverge exponentially.
- **If $\lambda_{\max} = \infty$:** The system is dominated by noise and exhibits unbounded instability.

6.6 Lyapunov Exponents: A Global Stability Measure

Why Use Lyapunov Exponents and how are they useful?

Lyapunov exponents address the classical stability analysis limitations by providing a *global*, time-dependent measure of how sensitive a system is to initial conditions, by quantifying the average exponential rate at which nearby trajectories in phase space diverge.

The Lyapunov time τ is defined as the inverse of the maximum Lyapunov exponent:

$$\tau = \frac{1}{\lambda_{\max}}$$

It represents the **characteristic time** over which initially close trajectories diverge by a factor of e . In practical terms, it sets the **predictability horizon**—beyond which even tiny uncertainties in initial conditions lead to drastically different outcomes even in deterministic systems.

This is crucial for real-world systems like weather forecasting or orbital mechanics, where deterministic equations govern evolution, but long-term predictions become impossible due to exponential sensitivity.

Lyapunov exponents extend the classical idea of stability from fixed points to entire trajectories. They allow us to:

- Diagnose chaotic behaviour.
- Quantify how fast predictability is lost.
- Compare global stability across different dynamical systems.

By capturing both the *magnitude* and *rate* of trajectory divergence, they provide a powerful lens to analyze complex, nonlinear, and chaotic systems beyond the reach of traditional stability theory.



6.6.1 Poincaré Section:

The solution of dynamic systems is often studied using the trajectories of flow in phase space. A Poincaré section is **a cross-section of phase space taken at discrete times**. It reduces a continuous flow to a discrete time map, making it easier to identify periodic, quasi-periodic, and chaotic regimes by observing intersections of trajectories. Note that the time interval from point to point need not be constant always. We expect some geometric properties of the flow and the Poincaré section to be the same:

- Dissipation $\rightarrow$ areas in the Poincaré section should contract.
- If the flow has an attractor, we should see it in the Poincaré section. It provides a means to visualize an otherwise messy, possibly aperiodic, attractor.

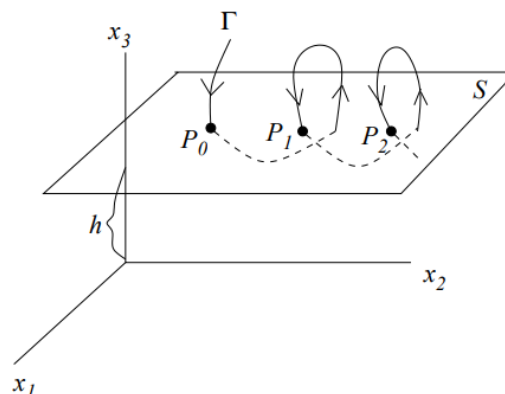


Figure 11: 2-D Poincaré section

In the figure (10), the points P_0, P_1, P_2 form the 2-D Poincaré section.

1. **Periodic** (flow is a closed orbit - eg., limit cycle)
2. **Quasi-periodic** (flow is a 3D torus with two fundamental frequencies f_1 and f_2 .)
 - **Irrational f_1/f_2 :** frequencies are incommensurate. The closed curve C appears continuous.
 - **Rational f_1/f_2 :** f_1 and f_2 are frequency locked. There are finite number of intersections along the curve C .
3. **Aperiodic flows:** not a simple curve, often a point cloud (statistical white noise). Deterministic aperiodic systems often display more order - mild departures from a simple curve.

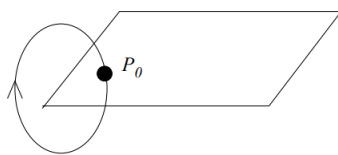


Figure 12: Periodic

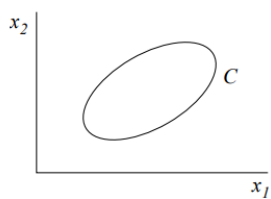


Figure 13: Incommensurate quasi periodic

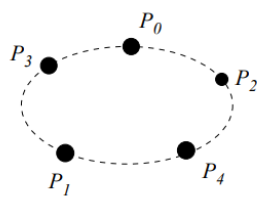


Figure 14: Frequency locked quasi periodic



Figure 15: Aperiodic point cloud

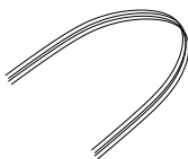


Figure 16: Deterministic aperiodic



7 Summary of Paper - Dynamics of a driven spheroid in a slow oscillating creeping shear flow

7.1 Important Terminologies

1. **Steady Flow:** A flow where velocity gradients do not change with time.
2. **Stokes Number:** Dimensionless parameter defined as:

$$\text{Stk} = \frac{t_0 u_0}{L_0}$$

where t_0 is the particle relaxation time, u_0 the flow velocity, and L_0 a characteristic length scale.

3. **Stokes Flow:** Flow at low Reynolds number ($Re < 1$). Drag force is linear with velocity.
4. **Navier-Stokes Equation (Incompressible):**

$$\rho \left(\frac{\partial \vec{u}}{\partial t} + (\vec{u} \cdot \nabla) \vec{u} \right) = -\nabla p + \mu \nabla^2 \vec{u} + \vec{f}$$

5. Types of Fluids:

- *Newtonian:* Constant viscosity.
- *Non-Newtonian:* Viscosity varies with applied shear stress.
- *Pseudoplastics:* Viscosity decreases with shear rate.
- *Dilatant:* Viscosity increases with shear rate.

6. Modelling Non-Newtonian fluids:

- *Power law model:* Generalized model for all
- *Bird-Carreau model:* valid in very low and high shear rates too.
- *Herschel-Belkley model:* combines behaviour of Bingham plastics and power-law fluids

7. **State Space:** The set of all possible configurations of a dynamical system.
8. **Gimbal Lock:** Loss of one degree of freedom in 3D rotation.
9. **Jeffery's Orbits:** Motion of a spheroid in viscous shear flow showing periodic orbits.



10. **Strouhal number:** It is a dimensionless parameter that describes the oscillating flow mechanism and represents the ratio of vibration-to-flow velocities or the ratio of local force to inertia force in non-stationary processes.

7.2 Paper Summary

- Studies orientation dynamics of a sinusoidally forced spheroid in weak/strong oscillatory shear.
- Governing equations reduce to Jeffery's orbits in absence of forces.
- Explores different dynamic regimes: quiescent, simple shear, weak and strong oscillatory shear in the absence of inertial forces.
- Uses Poincaré maps, phase plots, and Lyapunov exponents for analysis.
- Finds shape-based particle separation is possible using flow and forcing variations.

7.3 Introduction

The present work investigates the rotational evolution of a single spheroidal particle in a slow and weak/strong oscillating shear flow, assuming a dilute suspension regime to neglect particle–particle interactions. Such analysis is crucial, as orientation impacts key physical quantities like drag, lift, and rheological behaviour of suspensions, with practical implications in drug delivery, colloidal separation, and fluidized beds. We derive the governing equations using a Lagrangian approach, both in Cartesian and spherical coordinates, and solve them numerically across different parametric regimes. Through tools such as Poincaré maps, phase diagrams, and Lyapunov exponents, we analyze regular and chaotic dynamics and compare them with the classical Jeffery case. Ultimately, the study aims to deepen our understanding of how oscillatory flow modifies particle rotation and how this can be harnessed for shape-based particle separation in practical systems.

7.4 System Setup and Modelling

- Single spheroid in 1D oscillating shear.
- Low-Reynolds number (Stokes flow), neglecting inertia and Brownian motion.
- Fixed parameters: $\omega_2 = 1.0$, $a \in \{0.1, 1.0\}$, $\Omega_1 \in \{0.0, 0.1\}$
- External torque: $\vec{k} = (0, k_2, 0)$, with $k_2 \in [0, 15]$
- Aspect ratio $r_e \in [0.2, 2.0]$ to cover both oblate and prolate shapes

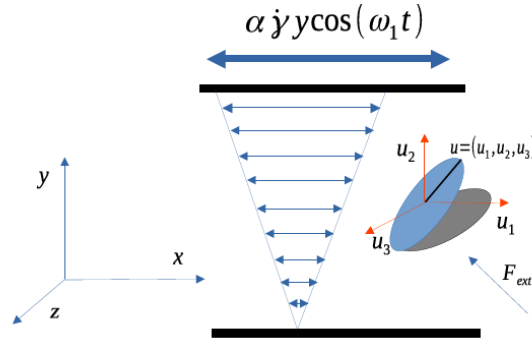


Figure 17: A Schematic representation of a spheroid suspended in an oscillating shear flow under the action of an external force.

- The dilute regime assumption ensures that inter-particle interactions are negligible. In this regime, particles rotate freely with three degrees of freedom without being influenced by neighbours. The volume fraction Ψ of a spheroid suspended in a fluid of volume V is given by:

$$\Psi = \frac{4\pi ab^2}{3Vd}, \quad (8)$$

where:

- a is the polar (major) radius,
- b is the equatorial (minor) radius,
- $d > 1$ defines the degree of dilution.

The aspect ratio of the spheroid is defined as:

$$r_e = \frac{a}{b}. \quad (9)$$

Under the dilute regime, the following conditions must be satisfied:

- **Prolate spheroid** ($r_e > 1$):

The particle volume is $V = (2a)^3$. The dilution condition becomes:

$$\Psi \cdot r_e^2 \cdot d < 1.$$

- **Oblate spheroid** ($r_e < 1$):

The particle volume is $V = (2b)^3$. The dilution condition becomes:

$$\Psi \cdot d/r_e < 1.$$



7.5 Torque and Governing Equations

We begin with Brenner's model for the torque $\mathbf{L}$ acting on a small spheroidal particle immersed in a linear viscous flow. The torque results from the external forcing as well as the surrounding fluid (viscous resistance). The general form of the torque is given as:

$$\mathbf{L} = \text{External Torque} + \text{Viscous Torque}. \quad (10)$$

In a simple linear flow field, Brenner's expression for the torque on a particle with fore-aft symmetry is:

$$\mathbf{L} = \mu [\mathbf{r}_k \cdot (\boldsymbol{\omega} - \boldsymbol{\Omega}) + \hat{\mathbf{s}} : \mathbf{S}], \quad (11)$$

where:

- μ is the dynamic viscosity of the fluid,
- $\boldsymbol{\omega}$ is the particle angular velocity,
- $\boldsymbol{\Omega} = \frac{1}{2} \nabla \times \mathbf{V}$ is the fluid vorticity,
- $\mathbf{S} = \frac{1}{2} (\nabla \mathbf{V} + (\nabla \mathbf{V})^T)$ is the rate-of-strain tensor,
- $\mathbf{r}_k$ and $\hat{\mathbf{s}}$ are geometric tensors depending on the particle shape and surface.

$\boldsymbol{\omega} - \boldsymbol{\Omega}$: Rotational slip velocity—how much the particle's rotation differs from local fluid vorticity. $\hat{\mathbf{s}} : \mathbf{S}$: Measures how fluid deformation (strain) contributes to torque due to particle shape.

A time-dependent external torque is also considered, representing a magnetic or electric field acting on a dipolar particle:

$$\mathbf{L}_{\text{ext}} = \mathbf{k} \times \mathbf{u} \cos(\omega_2 t), \quad (12)$$

where $\mathbf{k} = (k_1, k_2, k_3)$ is the torque strength vector, ω_2 is the forcing frequency, and $\mathbf{u}$ is the particle orientation unit vector.

7.6 Derivation Summary

1. Model the fluid flow as a one-dimensional oscillating shear:

$$\mathbf{V} = a \dot{\gamma} y \cos(\omega_1 t) \mathbf{e}_x,$$

where a is the amplitude and ω_1 is the frequency of the fluid oscillations.

2. Determine the local vorticity $\boldsymbol{\Omega}$ and rate-of-strain tensor $\mathbf{S}$.



3. Use torque balance and rotational slip to relate torque to the particle's angular velocity $\boldsymbol{\omega}$.
4. Write the particle orientation dynamics via:

$$\dot{\mathbf{u}} = \boldsymbol{\omega} \times \mathbf{u}.$$

5. Substitute all expressions and simplify into coupled non-linear differential equations.

Final Differential Equations

In Cartesian Coordinates Let $\mathbf{u} = (u_1, u_2, u_3)$, the orientation vector. The governing equations become:

$$\dot{u}_1 = Pu_2(1 - 2u_1^2)a \cos(\omega_1 t) + Qu_2a \cos(\omega_1 t) + Rb_1 \cos(\omega_2 t), \quad (13)$$

$$\dot{u}_2 = Pu_1(1 - 2u_2^2)a \cos(\omega_1 t) - Qu_1a \cos(\omega_1 t) + Rb_2 \cos(\omega_2 t), \quad (14)$$

$$\dot{u}_3 = -2Pu_1u_2u_3a \cos(\omega_1 t) + Rb_3 \cos(\omega_2 t), \quad (15)$$

where:

$$b_1 = u_3(u_3k_1 - u_1k_3) - u_2(u_1k_2 - u_2k_1),$$

$$b_2 = u_1(u_1k_2 - u_2k_1) - u_3(u_2k_3 - u_3k_2),$$

$$b_3 = u_2(u_2k_3 - u_3k_2) - u_1(u_3k_1 - u_1k_3).$$

In Spherical Coordinates, Let θ be the azimuthal angle and ϕ the polar angle. To remove the singularity at $\theta = 0$, define $\phi_1 = \phi \sin \theta$, and the equations become:

$$\dot{\theta} = \frac{P}{2} \sin(2\theta) \sin(2\phi)a \cos(\omega_1 t) + R [\cos \theta \cos \phi k_1 + \cos \theta \sin \phi k_2 - \sin \theta k_3] \cos(\omega_2 t), \quad (16)$$

$$\dot{\phi}_1 = \frac{P}{2} \phi \cos \theta \sin(2\theta) \sin(2\phi)a \cos(\omega_1 t) + \sin \theta (P \cos(2\phi) - Q)a \cos(\omega_1 t) \quad (17)$$

$$+ R(-\sin \phi k_1 + \cos \phi k_2) + \phi(\cos^2 \theta \cos \phi k_1 + \cos^2 \theta \sin \phi k_2 - \sin \theta \cos \theta k_3) \cos(\omega_2 t). \quad (18)$$

$$P = \frac{r_e^2 - 1}{2(r_e^2 + 1)},$$

$$Q = \frac{1}{2},$$

$$R = \frac{3r_e^4 b(2r_e^2 - 1) - 1}{16\pi(r_e^2 - 1)(r_e^2 + 1)},$$



where

$$b = \begin{cases} \frac{\ln(r_e + \sqrt{r_e^2 - 1})}{r_e \sqrt{r_e^2 - 1}}, & r_e > 1 \quad (\text{prolate}) \\ \frac{\cos^{-1}(r_e)}{r_e \sqrt{1 - r_e^2}}, & r_e < 1 \quad (\text{oblate}) \end{cases}$$

These equations fully describe the non-linear orientation dynamics of a spheroid in an oscillating shear flow subject to an external periodic torque.

7.7 Numerical Method

- Runge-Kutta 4th Order
- Adaptive time-step in MATLAB

7.8 Key Findings

Regimes of External Force Leading to Chaos

1. Simple Shear Flow (No Oscillations)

- Reproduced results from Kumar *et al.* for validation.
- Exhibited multiple windows of chaotic behaviour for both prolate and oblate spheroids.
- Example for prolate spheroid with aspect ratio $r_e = 1.6$:
Chaotic regimes observed for k_2 in the intervals:

$$(0.01, 4.58), \quad (5.2, 8.06), \quad (11.67, 12.45).$$

2. Weak Oscillating Shear ($\alpha = 0.1, \omega_1 = 0.1$)

- Presence of a single continuous chaotic regime.
- Example for $r_e = 1.6$:
Chaotic dynamics observed for $k_2 \in [5.27, 15.0]$; below $k_2 = 5.27$ the behaviour was regular.

3. Strong Oscillating Shear ($\alpha = 1.0, \omega_1 = 0.1$)

- Contrary to expectation, strong oscillations suppressed chaos and promoted regular dynamics.



- Key insight: Increased shear strength does not necessarily lead to more complex or chaotic behaviour.

4. Oblate Spheroids ($r_e < 1$)

- Showed lower sensitivity to external forcing in terms of chaotic behaviour compared to prolate spheroids.
- For $r_e = 0.6$, the chaotic behaviour observed in simple shear was entirely regularized under both weak and strong oscillatory shear.
- Exception: $r_e = 0.8$ exhibited *non-monotonic* response — alternating regular and chaotic sub-regimes.

Implications

- The combination of external torque, flow strength, and oscillation frequency significantly influences the complexity of particle orientation dynamics.
- Even weak oscillations in the fluid flow can act as effective control mechanisms to suppress chaotic regimes.
- These findings can be strategically employed in designing suspension-based separation techniques by tuning dynamical regimes based on particle shape and applied flow parameters.

7.9 Analysis of Orientation Dynamics

1. Poincaré Sections

- **Prolate Spheroids:**
 - Under weak and slow oscillating shear, orientation trajectories produced scattered and dense Poincaré points, indicating **chaotic behaviour**.
 - Under simple shear (both weak and strong), Poincaré sections shrank into discrete loops, reflecting **regularization of dynamics**.
 - Under strong oscillating shear, periodic patterns emerged in the Poincaré sections, indicating a return to **periodic behaviour**.
- **Oblate Spheroids:**
 - In simple shear, Poincaré sections appeared as thick curves, indicating **mildly regular** behaviour.



- Under slow oscillating flow (both weak and strong), Poincaré sections collapsed into discrete points, showing **regularization**.

2. Trajectory Attractors in (u_1, u_2, u_3) Space

- In strong oscillating shear flows, attractors flattened into 2D surfaces within the flow-gradient plane, indicating **planar confinement**.
- This confinement was **not observed** under weak oscillations, where motion remained **three-dimensional and irregular**.

3. Time Series Plots

- Components u_1 and u_2 showed increased oscillation amplitude with increasing flow strength, for both oblate and prolate spheroids.
- The component u_3 showed rapid decay in oscillatory flows, implying **limited out-of-plane rotation** under strong oscillations.
- Prolate spheroids exhibited higher amplitude and more complex oscillations in all directions compared to oblate spheroids.

4. Lyapunov Exponents

- Computed numerically to confirm the qualitative behavior seen in simulations.
- Results showed that **slow oscillatory shear** drives the system toward **regular dynamics**, with dominant exponents ≤ 0 .
- This supports and validates the observations from Poincaré sections, trajectory attractors, and time series.



Aspect ratio	Shear strength	Shear	
		Simple	Oscillating
$r_e = 0.2$	$\alpha = 0.1$	0.011	-0.014
	$\alpha = 1.0$	0.006	1.18×10^{-4}
$r_e = 0.4$	$\alpha = 0.1$	0.002	-0.014
	$\alpha = 1.0$	0.407	-
$r_e = 0.6$	$\alpha = 0.1$	0.085	0.014
	$\alpha = 1.0$	0.702	-1.4×10^{-15}
$r_e = 0.8$	$\alpha = 0.1$	0.290	0.029
	$\alpha = 1.0$	0.212	1.36×10^{-17}
$r_e = 1.2$	$\alpha = 0.1$	0.339	0.063
	$\alpha = 1.0$	0.199	-3.66×10^{-16}
$r_e = 1.4$	$\alpha = 0.1$	0.328	0.277
	$\alpha = 1.0$	0.044	7.87×10^{-16}
$r_e = 1.6$	$\alpha = 0.1$	0.277	0.381
	$\alpha = 1.0$	0.098	-1.21×10^{-16}
$r_e = 1.8$	$\alpha = 0.1$	0.177	0.116
	$\alpha = 1.0$	0.217	1.36×10^{-17}
$r_e = 2.0$	$\alpha = 0.1$	0.456	0.015
	$\alpha = 1.0$	0.235	-5.61×10^{-16}

Figure 18: Table showing Lyapunov exponents of trajectories of spheroid dynamics suspended in weak and strong flows with either slow or no phase oscillations for different particle shapes

The Poincaré sections, phase space plots for u_1, u_2 and u_3 and 3D attractor trajectory for prolate and oblate spheroids under various shear force and oscillatory conditions simulated in MATLAB have been attached in the supplementary report.

7.10 Particle Separation

The observations of the existence of multiple alternate chaotic and regular regimes, and sensitivity of the dynamics to particle aspect ratio and flow frequency, can be used to devise a better particle separation mechanism for suspensions having desired properties.

Proposed Mechanism: Combine simple shear with superimposed slow oscillations to selectively destabilize or regularize particles based on aspect ratio. This makes it possible to isolate one type of spheroid (e.g., prolate) from a mixed suspension based on their unique dynamical signature.



8 Harmonic Balance Method

8.1 Does HBM Lose Non-linear Behaviour? When do we use HBM?

The Harmonic Balance Method (HBM) *does* capture non-linearities—but only their *periodic* part.

What it captures:

- Non-linear interactions between harmonics (i.e., mode coupling).
- The shape of the steady-state periodic solution of a non-linear system.
- Non-sinusoidal waveform distortions caused by non-linear effects.

What it doesn't capture:

- Transient dynamics (HBM solves directly for the steady-state).
- Chaotic behaviour or non-periodic attractors (unless extended with methods like multi-time scales or continuation).
- Sharp transitions or bifurcations—unless many harmonics are used to resolve them.

When is HBM best suited?

- When the system settles into a periodic motion, even if it is non-linear and strongly distorted.
- When your primary interest is in periodic steady-state responses, not transient dynamics.

In the simulations below for studying chaotic dynamics, we must note that sometimes the Harmonic Balance Method may have converged to a stable periodic solution, possibly missing chaotic attractors due to:

- The choice of initial guess leading to convergence in the basin of attraction of a stable orbit,
- The number of harmonics being insufficient to capture high-frequency chaotic components,
- The inherent filtering effect of HBM favouring regular, steady-state periodic solutions.



8.2 Interpretation of Phase Space and Orientation Trajectory after applying HBM to eqns (16) and (17)

From the obtained phase space plots of u_1 , u_2 , and u_3 , and the corresponding 3D orientation trajectory $(u_1(t), u_2(t), u_3(t))$, we make the following observations and inferences:

Phase Space Analysis

Each phase space plot shows a closed or elliptical loop in the $(u_k, \dot{u}_k)$ plane for $k = 1, 2, 3$. These closed orbits are a strong indicator of periodic or quasi-periodic behaviour. Specifically:

- The absence of divergence or spreading in the phase plane suggests that the system has settled into a bounded, stable orbit.
- The slight presence of nested loops (particularly in u_1 and u_3) indicates the contribution of multiple harmonics, which is consistent with the use of a truncated Fourier series in the Harmonic Balance Method (HBM).

3D Trajectory Analysis

The 3D plot of the orientation vector (u_1, u_2, u_3) reveals a smooth, curved trajectory confined to a narrow region in space:

- The trajectory does not fill a volume or show irregular wandering, which suggests a low-dimensional attractor.
- The system appears to evolve along a stable periodic or quasi-periodic orbit, without signs of chaos such as irregularity or sensitivity to initial conditions.



9 Credits for images and References

- Duffing Oscillator
- Newton-Raphson method
- Medium article on heat equation
- Euler's formula
- Wikipedia - Duffing Oscillator
- Lyapunov Exponent
- MIT-OpenCourseWare - Poincaré section
- Wikipedia - Phase portrait

10 Bibliography

1. Dynamics of a driven spheroid in a slow oscillating creeping shear flow
2. Jeffery, George Barker. "The motion of ellipsoidal particles immersed in a viscous fluid." *Proceedings of the Royal Society of London. Series A, Containing papers of a mathematical and physical character* 102.715 (1922): 161-179.
3. Orthogonality of Sinusoidal functions
4. Medium article on norms
5. Steven Strogatz - Non-Linear Dynamics and Chaos
6. Blake Pi — Medium Article on Chaos and Determinism.
7. Wikipedia - Lyapunov Exponents
8. Wikipedia - Lyapunov Time
9. Wikipedia - Van der Pol oscillator
10. Wikipedia - Phase portrait
11. Non-linear least squares